

STAT 153 & 248 - Time Series

Lecture Six

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Aditya Guntuboyina

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1 Nonlinear Regression

Our next topic is nonlinear regression models. In these models, certain parameters appear in a nonlinear fashion. One simple example of such a model is:

$$y_t = \beta_0 + \beta_1 t + \beta_2 \text{ReLU}(t - c) + \epsilon_t \quad (1)$$

with $\epsilon_t \stackrel{\text{i.i.d.}}{\sim} N(0, \sigma^2)$. Here $\text{ReLU}(t - c) = (t - c)_+$ equals 0 if $t \leq c$ and equals $t - c$ if $t \geq c$. We can also write

$$\text{ReLU}(t - c) = (t - c)_+ = (t - c)I\{t > c\} = \max(t - c, 0).$$

$(\cdot)_+$ is also called the positive part function, or, the ramp function.

The model (1) says that for times $t \leq c$, the slope of the regression line is β_1 , while for $t > c$, the slope changes to $(\beta_1 + \beta_2)$. We shall refer to (1) as the 'Change of Slope' model. An alternative name for this model is "Broken-stick regression". This is because the function

$$t \mapsto \beta_0 + \beta_1 t + \beta_2 \text{ReLU}(t - c)$$

resembles a broken stick.

The unknown parameters for this model are $c, \beta_0, \beta_1, \beta_2$ as well as σ . The unknown parameter c makes (1) a nonlinear regression model. If c were known, then (1) would be a linear regression model:

$$y = X_c \beta + \epsilon \quad (2)$$

with

$$X_c = \begin{pmatrix} 1 & 1 & \text{ReLU}(1 - c) \\ 1 & 2 & \text{ReLU}(2 - c) \\ 1 & 3 & \text{ReLU}(3 - c) \\ \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot \\ 1 & n & \text{ReLU}(n - c) \end{pmatrix} \quad \text{and } \beta := \begin{pmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \end{pmatrix} \quad \text{and } \epsilon = \begin{pmatrix} \epsilon_1 \\ \epsilon_2 \\ \cdot \\ \cdot \\ \epsilon_n \end{pmatrix}$$

1.1 Parameter Estimation

Least squares again is the most basic estimation procedure. The sum of squares is:

$$S(\beta_0, \beta_1, \beta_2, c) := \sum_{t=1}^n (y_t - \beta_0 - \beta_1 t - \beta_2 \text{ReLU}(t - c))^2. \quad (3)$$

We need to minimize this over all the four variables $\beta_0, \beta_1, \beta_2, c$. Using matrix notation, we can write

$$S(\beta, c) = \|y - X_c \beta\|^2.$$

If we fix c , then it is easy to minimize $S(\beta, c)$ over β . This is the same as linear regression and the minimizing β is given by:

$$\hat{\beta}(c) := (X_c^T X_c)^{-1} X_c^T y, \quad (4)$$

and the smallest value of $S(\beta, c)$ for fixed c is $S(\hat{\beta}(c), c)$ which is just the RSS in the multiple linear regression with fixed c . We use the notation:

$$RSS(c) = S(\hat{\beta}(c), c) = \min_{\beta} S(\beta, c). \quad (5)$$

The least squares estimates of β and c can therefore be found in the following way:

1. Fix a finite set of possible values of c . In this change of slope model with $t = 1, \dots, n$, it is reasonable to take a fine grid of values in the range $(1, n)$. This range makes sense because of the following. If $c \geq n$, then $\text{ReLU}(t - c) = 0$ for all $t = 1, \dots, n$ so the term $\text{ReLU}(t - c)$ is not needed in the model (1). If $c \leq 1$, then $\text{ReLU}(t - c) = t - c$ for all $t = 1, \dots, n$ so that the ReLU term in (1) becomes linear so the model again becomes a linear function of t .
2. For each value of c in the chosen set, calculate $\hat{\beta}(c)$ and define $RSS(c) = S(\hat{\beta}(c), c)$.
3. Take $\hat{c}$ to be the value of c which minimizes $RSS(c)$.
4. Take $\beta = \hat{\beta}(\hat{c})$.

1.2 Bayesian Inference

For uncertainty quantification, let us consider Bayesian inference for this model. We need to first write down the likelihood and prior, and then use them to derive the posterior. For the likelihood, we shall assume (just as in usual linear regression) that the normal distribution for the errors $\{\epsilon_t\}$ in (1): $\epsilon_t \stackrel{\text{i.i.d.}}{\sim} N(0, \sigma^2)$. This leads to the likelihood:

$$\prod_{t=1}^n \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(y_t - \beta_0 - \beta_1 t - \beta_2(t - c)_+)^2}{2\sigma^2}\right) = (2\pi)^{-n/2} \sigma^{-n} \exp\left(-\frac{S(\beta, c)}{2\sigma^2}\right) \quad (6)$$

where again $S(\beta, c)$ is the sum of squares (3).

For the prior, recall that, for linear regression, we used:

$$\beta_0, \beta_1, \beta_2, \log \sigma \stackrel{\text{i.i.d.}}{\sim} \text{unif}(-\infty, \infty).$$

Here $\text{unif}(-\infty, \infty)$ can be thought of as short form for $\text{unif}(-C, C)$ for a very large constant C . We will use the same prior here also. However we also need to supply a prior for the

parameter c . As mentioned above, we will restrict c to be in the range $(1, n)$ (open interval from 1 to n). Because $c \in (1, n)$, a natural prior is to take:

$$c \sim \text{uniform}(1, n).$$

In practice, we would not be interested in values of c that are near the boundaries 1 and n . This is because if c is very close to 1 or n , then the model is essentially linear for most of dataset. To reflect this, we can take the prior for c to be uniform in smaller range than $(1, n)$ ignoring points near each end; for example, uniform on $(5, n - 5)$. More generally, let R be a closed interval strictly contained in $(1, n)$ and we shall take

$$c \sim \text{uniform}(R) \tag{7}$$

As a concrete example, we can take $R = [5, n - 5]$.

The posterior joint density of all the parameters β, c, σ (here β denotes the vector consisting of $\beta_0, \beta_1, \beta_2$) is given by

$$\text{posterior}(\beta, c, \sigma) \propto \text{likelihood} \times \text{prior}.$$

The likelihood is given in (6) and the prior is

$$\begin{aligned} \text{prior density} &= \frac{I\{-C < \beta_0, \beta_1, \beta_2, \log \sigma < C\} I\{c \in R\}}{(2C)^4 \sigma |R|} \\ &\propto \frac{I\{-C < \beta_0, \beta_1, \beta_2, \log \sigma < C, c \in R\}}{\sigma}. \end{aligned}$$

Above, $|R|$ denotes the length of the region R . We have dropped in the next step because this length is a constant (that does not depend on the actual value of c).

We shall drop the indicator terms involving C because C is very large (think ∞). This will lead to

$$\text{prior density} \propto \frac{I\{\sigma > 0, c \in R\}}{\sigma}$$

The posterior is then given by

$$\text{posterior}(\beta, c, \sigma) \propto \sigma^{-n-1} \exp\left(-\frac{S(\beta, c)}{2\sigma^2}\right) I\{\sigma > 0\} I\{c \in R\}.$$

To get the posterior density of c alone (c is the most important parameter in the model (1)), we need to integrate the joint posterior density above with respect to β and σ :

$$\text{posterior}(c) \propto I\{c \in R\} \int_0^\infty \sigma^{-n-1} \int \exp\left(-\frac{S(\beta, c)}{2\sigma^2}\right) d\beta d\sigma.$$

Let us first calculate the inner integral. $S(\beta, c)$ is a quadratic function in β so that $\int \exp(-S(\beta, c)/(2\sigma^2)) d\beta$ should be related to the normalizing constants in the multivariate normal density. To figure the integral precisely, we first use the Pythagorean identity (discussed previously):

$$\begin{aligned} S(\beta, c) &= S(\hat{\beta}(c), c) + \left(\beta - \hat{\beta}(c)\right)^T X_c^T X_c \left(\beta - \hat{\beta}(c)\right) \\ &= RSS(c) + \left(\beta - \hat{\beta}(c)\right)^T X_c^T X_c \left(\beta - \hat{\beta}(c)\right) \end{aligned}$$

where $\hat{\beta}_c$ is given in (4). The value $S(\hat{\beta}_c, c)$ is equal to $RSS(c)$ as noted in (5). Thus

$$\begin{aligned}
\int \exp\left(-\frac{S(\beta, c)}{2\sigma^2}\right) d\beta &= \int \exp\left(-\frac{RSS(c)}{2\sigma^2}\right) \exp\left(-\frac{(\beta - \hat{\beta}(c))^T X_c^T X_c (\beta - \hat{\beta}(c))}{2\sigma^2}\right) d\beta \\
&= \exp\left(-\frac{RSS(c)}{2\sigma^2}\right) \int \exp\left(-\frac{(\beta - \hat{\beta}(c))^T X_c^T X_c (\beta - \hat{\beta}(c))}{2\sigma^2}\right) d\beta \\
&= \exp\left(-\frac{RSS(c)}{2\sigma^2}\right) (\sqrt{2\pi})^p \sqrt{\det(\sigma^2(X_c^T X_c)^{-1})} \\
&= \exp\left(-\frac{RSS(c)}{2\sigma^2}\right) (\sqrt{2\pi})^p \sigma^p |X_c^T X_c|^{-1/2}
\end{aligned}$$

where $|X_c^T X_c| = \det(X_c^T X_c)$. Here $p = 3$ because there are three components inside β . Therefore

$$\begin{aligned}
\text{posterior}(c) &\propto I\{c \in R\} \int_0^\infty \sigma^{-n-1} \exp\left(-\frac{RSS(c)}{2\sigma^2}\right) (\sqrt{2\pi})^p \sigma^p |X_c^T X_c|^{-1/2} d\sigma \\
&\propto I\{c \in R\} |X_c^T X_c|^{-1/2} \int_0^\infty \sigma^{-n+p-1} \exp\left(-\frac{RSS(c)}{2\sigma^2}\right) d\sigma.
\end{aligned}$$

The change of variable $\sigma = s\sqrt{RSS(c)}$, gives

$$\begin{aligned}
\text{posterior}(c) &\propto I\{c \in R\} |X_c^T X_c|^{-1/2} \left(\frac{1}{RSS(c)}\right)^{(n-p)/2} \int_0^\infty s^{-n+p-1} \exp\left(-\frac{1}{2s^2}\right) dt \\
&\propto I\{c \in R\} |X_c^T X_c|^{-1/2} \left(\frac{1}{RSS(c)}\right)^{(n-p)/2}.
\end{aligned}$$

This posterior will be evaluated numerically over a grid of values of c in the range R . The term $|X_c^T X_c|^{-1/2}$ will become infinite when $|X_c^T X_c| = 0$ i.e., when X_c does not have full column rank. This is the case when c is outside of the range $(1, n)$. When c is in the range $(1, n)$, the determinant of $X_c^T X_c$ will be non-zero but it will still be good to not consider c too close to 1 or n to prevent numerical instability due to near-singularity. This is why it is good to take R to be a region of the form $[5, n-5]$ so it is not too close to the endpoints 1 and n .

Let $\hat{c}$ denote the least squares estimate of c . Note that this is the minimizer of $RSS(c)$ over all values of c . We can then rewrite the above posterior formula as:

$$\text{posterior}(c) \propto I\{c \in R\} |X_c^T X_c|^{-1/2} \left(\frac{RSS(\hat{c})}{RSS(c)}\right)^{(n-p)/2}. \quad (8)$$

The main term in the posterior (8) is $(RSS(\hat{c})/RSS(c))^{(n-p)/2}$. This term takes its largest value when c equals the least squares estimator $\hat{c}$ (note $\hat{c}$ minimizes $RSS(c)$) (note that the region R will usually be such that $\hat{c} \in R$). The size of the power $n - p$ determines the amount of concentration of $(RSS(\hat{c})/RSS(c))^{(n-p)/2}$ around $\hat{c}$. When $n - p$ is large, $(RSS(\hat{c})/RSS(c))^{(n-p)/2}$ is very tightly concentrated around $\hat{c}$. The other term $|X_c^T X_c|^{-1/2}$ generally does not vary significantly with c for small regions around $\hat{c}$. This implies that the overall posterior is usually (especially when $n - p$ is large) tightly concentrated around $\hat{c}$.